

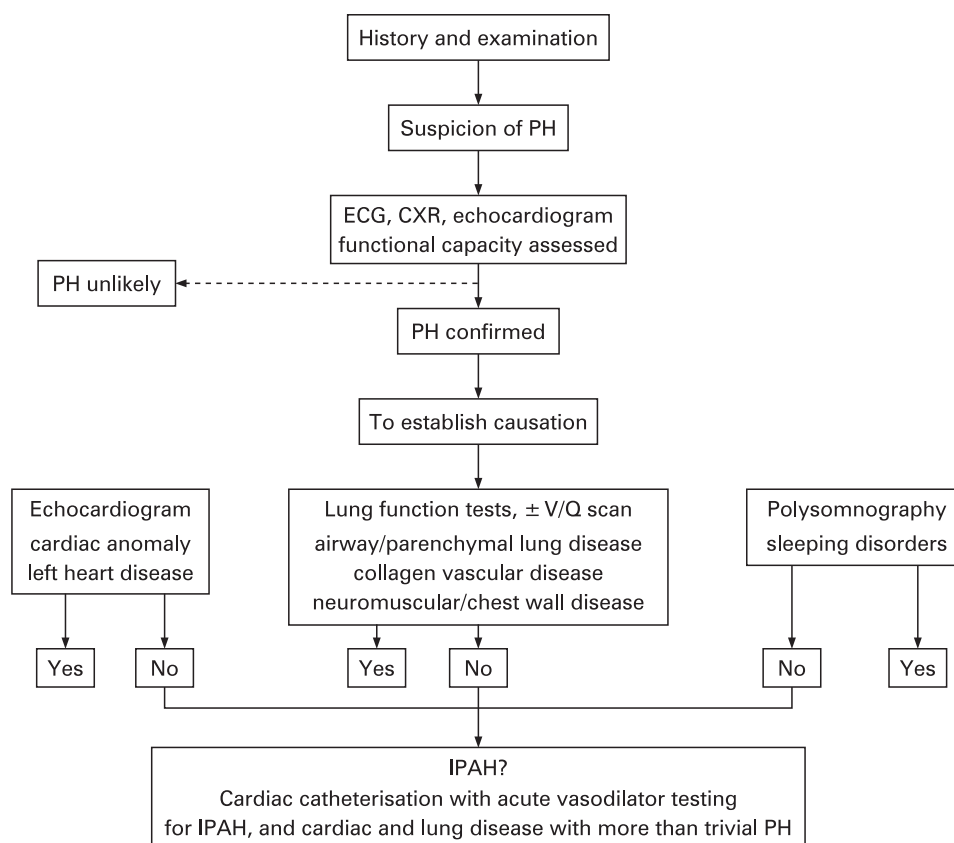


Figure 1 Investigating pulmonary hypertension: diagnostic strategy. CXR, chest x ray; PH, pulmonary hypertension; V/Q, ventilation/perfusion.

APAH

Successful treatment of the pulmonary hypertension depends on the rapidity with which the physician treating the underlying disease recognises that pulmonary hypertension is a major complicating factor in the clinical picture.

Presentation and the investigation of children with APAH are determined by the underlying disease process. The principle aetiologies are as follows.

Chronic hypoxia

Chronic hypoxia leads to pulmonary hypertension. Chronic lung disease of prematurity is a relatively common cause of moderate pulmonary hypertension. Interstitial lung disease, as strictly defined, is uncommon in children, as are the pulmonary complications of connective tissue disease. When pulmonary hypertension does occur in association with connective tissue disease in children, it is usually severe. Of the developmental abnormalities, the commonest association is congenital diaphragmatic hernia. In these patients pulmonary hypertension is relatively common in the neonatal period. It usually abates but can persist indefinitely, be severe and require long term treatment. The extent to which lung disease is causing or contributing to an elevated PAP is revealed by pulmonary function tests, the blood gases, ventilation/perfusion and high resolution CT scans.

Persistent pulmonary hypertension of the newborn (PPHN)

This condition is usually self-limiting unless there is a significant irreversible underlying problem such as chronic lung disease of prematurity. Alveolar hypoplasia, usually

accompanied by a degree of dysplasia, is signalled by persistent failure to wean off the ventilator and an open lung biopsy may be needed to guide management. PPHN may also be the first indication of IPAH.

Congenital heart disease

Sustained pulmonary hypertension is sometimes recognised following a technically successful intra-cardiac repair and can be severe. Typically, the child improves clinically after the repair but then becomes short of breath and tires easily. The more severely affected children are investigated and treated as though they had IPAH. Classical Eisenmenger syndrome in the patient who initially had a large left to right shunt and then slowly developed pulmonary vascular disease with shunt reversal, central cyanosis and clubbing is not usually a problem in childhood. Most patients begin to deteriorate in their teenage years. Any child who appears to have Eisenmenger syndrome should be evaluated carefully to ensure that the cardiac diagnosis is correct and complete and that remedial surgery is not feasible. Sleep studies may be indicated in those with trisomy 21. Paediatric cardiology units are familiar with the small group of children who had a high pulmonary vascular resistance when first seen and were therefore deemed inoperable. In such children the resistance probably remained high from birth. They frequently become symptomatic in childhood and may benefit from medical therapy. Complex surgically naive and post-operative cases for whom curative or palliative surgery is not feasible frequently need reappraisal including cardiac catheterisation with a view to palliation by medical therapy.